

Use Cases:

UC1: Power ON

UC2: Power OFF

UC3: Cardiac Arrhythmia Detection

UC4: Real-Time CPR Feedback

UC5: Shock Delivery

UC6: CPR and Post-Shock Care

Power On AED Plus (UC1)

Primary Actor: Device User

Preconditions:

- The first responder is trained in the use of the AED Plus.
- The first responder locates the AED Plus.
- The first responder identifies the need for an AED Plus.

Postconditions:

- The AED Plus is powered on and ready for use.
- The AED Plus has performed a self-test to ensure it's functioning properly.
- The AED Plus provides a visual and audible indication that it is operational.

Main Success Scenario:

- The first responder presses the power button on the AED Plus.
- The AED Plus powers on and begins to give audio instructions for use.
- The AED Plus performs a self-test to ensure it's functioning properly.
- The AED Plus provides a visual and audible indication that it is operational.

Extensions:

1. The AED Plus does not power on:
 - The device user checks the battery installation.
 - The device user replaces the battery if necessary.

Power Off AED (UC2)

Primary Actor: Device User

Preconditions:

- The AED is currently powered on.
- No immediate medical emergency is occurring.

Postconditions:

- The AED Plus is powered off.
- No further instructions or feedback are provided by the AED.

Main Success Scenario:

- The user determines the AED is no longer needed.
- The power button on the AED Plus is held down to initiate shutdown.
- The AED Plus powers off after confirming the user's intent.

Extensions:

1. If the AED Plus does not power off:
 - The user is prompted to check if an emergency procedure is in progress.
 - The user consults the manual for troubleshooting steps or contacts support.

Cardiac Arrhythmia Detection (UC3)

Primary actor: Device User

Preconditions:

- The AED is powered on.
- All self-tests are complete.
- The AED is in good physical condition.
- Electrodes are properly connected.

Postconditions:

- The patient's heart rhythm is analyzed.
- A determination is made whether a shock is needed.

Main Success Scenario:

- The electrodes are placed on the patients chest and upper body area as expressed in the instructions of use
- As the electrodes measure the heart rate of the patient, all nearby people including the person giving first aid are advised to not touch the patient
- Once the electrodes have gotten enough data regarding the heart rate of the patient then this data is sent back to the AED
- This is where it is processed within the AED and determined that the patient has Cardiac Arrhythmia
- The AED prepares to send a shock via the electrodes to the patient in order to restore their heart beat to a normal rhythm

Extensions:

1. If the device user cannot understand the instructions on the AED:
 - The AED will repeat the instructions.

- The AED will provide additional visual cues or prompts if available.
- 2. If the patient's heart rate is detected as bradycardia (below 60 beats per minute), tachycardia (above 100 beats per minute), or is inconsistent:
 - The AED will categorize the arrhythmia type and prepare for the appropriate response.
 - If the rhythm is not shockable, the AED will instruct the user to continue monitoring or start CPR.
- 3. If the AED detects that a shock is required:
 - The AED charges to the appropriate level for the first shock.
 - If the patient's heart rhythm does not stabilize after the first shock, the AED will charge for a subsequent shock and advise the user to deliver it.
 - The shock level may be adjusted according to the patient's response and AED protocols.
- 4. If the AED is unable to deliver a shock due to a malfunction or a safety concern:
 - The AED will alert the user of the issue.
 - The AED will advise the user to perform manual CPR and seek additional help.

Real-Time CPR Feedback (UC4)

Primary Actor: Device User

Preconditions:

- CPR is required to be performed on the patient.
- The AED simulation is operational and capable of providing CPR feedback.

Postconditions:

- Feedback on CPR performance is provided.

Success Guarantees:

- The simulation provides accurate, real-time feedback, improving the quality of CPR performed by the user.

Main Success Scenarios:

- The user begins performing CPR on the patient.
- The AED simulation monitors the rate and depth of chest compressions.
- The simulation provides real-time feedback, advising the user to adjust the rate or depth of compressions as necessary.
- The user adjusts CPR technique based on the feedback to ensure effective chest compressions.

Extensions:

1. CPR feedback system malfunctions:
 - The user is prompted to continue CPR without feedback.
 - The user checks electrode pad connections and device battery.
2. Feedback indicates inadequate CPR:
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 - The AED provides corrective verbal prompts.
3. Movement interferes with CPR quality monitoring:
 - The AED instructs the user to minimize movement.
4. Patient regains pulse during CPR:
 - The AED detects the change and advises to stop CPR.

Shock Delivery (UC5)

Primary Actor: Device User

Preconditions:

- The AED simulation is operational.
- A shockable heart rhythm has been detected.

Postconditions:

- The AED prompts the user to deliver a shock.
- The user delivers a shock.
- Post-shock instructions are provided.

Main Success Scenarios:

- The user is instructed to ensure no one is touching the patient.
- The user is prompted to press the shock button.
- Upon user action, the simulation mimics the delivery of an electrical shock.
- The AED provides post-shock instructions, typically advising the continuation or resumption of CPR.

Extensions:

1. AED fails to deliver shock:
 - The user checks the electrode connections and battery status.
 - The AED prompts to attempt redelivery or perform manual CPR.
2. Patient's condition changes before shock delivery:
 - The AED reevaluates the patient's rhythm and updates the treatment recommendation.
3. The user hesitates to deliver the shock:
 - The AED reissues the prompt to deliver the shock, emphasizing urgency.

CPR and Post-shock care (UC6)

Primary actor: Device user

Precondition:

- The AED is powered on and functional.
- Electrodes are properly connected.

Minimal guarantee: The AED should successfully read the patient's heart rhythm and issue the necessary amount of shock in order to help the patient, the AED also administers the appropriate shock level to the patient based on their condition

Main Success Scenario:

- After the patient has been shocked with the appropriate shock level required for the patient's condition
- After a shock is administered to the patient, the medical provider is required to start with chest compressions
- The medical provider is now required to begin CPR on the patient and to continue to monitor the patient's state of being until more professional medical personnel is available
- The instructions/steps to provide CPR are readily available on the AED for the medical provider
- The medical provider assisting the patient is required to tap the patient and speak very loudly to check if the patient is responsive and to regularly check in emotionally with the patient

Extensions:

1. If multiple shocks are needed:
 - CPR is performed immediately after each shock.
2. For CPR instructions:
 - The AED provides visual and audio CPR instructions.
3. If professional medical help is delayed:
 - Emergency services are called for assistance.

Use Case Diagram

